COMPARATIVE ANALYSIS OF DIFFERENT TYPES OF PROMPTING PATTERNS

Aim

To test and compare how different prompting patterns (naïve vs. basic prompts) affect the quality, accuracy, and depth of responses generated by ChatGPT across multiple scenarios.

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1. Explanation of Prompt Types

Naïve Prompt (Unstructured Prompt)

A naïve prompt is simple, vague, or lacks clear instructions. It does not provide enough context or constraints.

Example:

"Tell me about climate change."

Basic Prompt (Structured Prompt)

A basic prompt is clear, detailed, and structured. It includes context, format, and specific instructions.

Example:

"Explain climate change in 150 words, including its causes, effects, and one real-world example."

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2. Test Scenarios and Prompts

Scenario 1: Creative Story

- **Naïve Prompt:**
  "Write a story."
- **Basic Prompt:**
  "Write a short story (150–200 words) about a boy who discovers a hidden power. Include a twist ending and a moral."

Scenario 2: Factual Question

- **Naïve Prompt:**
  "What is AI?"
- **Basic Prompt:**
  "Define Artificial Intelligence in simple terms, explain its types, and give two real-world applications."

Scenario 3: Summarization

- **Naïve Prompt:**
  "Summarize photosynthesis."
- **Basic Prompt:**
  "Summarize the process of photosynthesis in 5 bullet points, including its importance and main steps."

Scenario 4: Advice / Recommendation

- **Naïve Prompt:**
  "How to study better?"
- **Basic Prompt:**
  "Give 5 effective study tips for college students, including time management and revision techniques."

Scenario 5: Problem Solving

- **Naïve Prompt:**
  "Solve this math problem."
- **Basic Prompt:**
  "Solve step-by-step: Find the value of x in the equation 2x + 5 = 15."

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3. Comparative Table of Results

Scenario	Prompt Type	Quality	Accuracy	Depth	Observation
Creative Story	Naïve	Medium	Medium	Low	Generic story, lacks creativity
Creative Story	Basic	High	High	High	Structured, engaging, includes twist
Factual Question	Naïve	Medium	Medium	Low	Basic definition only
Factual Question	Basic	High	High	High	Detailed explanation with examples
Summarization	Naïve	Medium	High	Low	Short but unstructured
Summarization	Basic	High	High	Medium	Clear bullet points, well-organized
Advice	Naïve	Medium	Medium	Low	General tips
Advice	Basic	High	High	High	Practical and structured advice
Problem Solving	Naïve	Low	Medium	Low	May skip steps

Scenario	Prompt Type	Quality	Accuracy	Depth	Observation
Problem Solving	Basic	High	High	High	Step-by-step clear solution

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4. Analysis

Quality

- Basic prompts produce more **organized and readable outputs**.
- Naïve prompts often lead to **generic or incomplete answers**.

Accuracy

- Accuracy is generally **better in basic prompts** due to clear instructions.
- Naïve prompts may miss key details.

Depth

- Basic prompts provide **detailed and comprehensive responses**.
- Naïve prompts tend to be **surface-level**.

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5. Key Observations

- ChatGPT performs **consistently better with structured prompts**.
- Naïve prompts sometimes work well for **simple or general tasks**.
- Complex tasks (problem solving, analysis) require **clear instructions**.

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6. Findings

- Prompt clarity directly impacts response quality.
- Adding constraints like:
    - Word limit
    - Format (bullet points, steps)
    - Specific requirements
      improves results significantly.

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7. Conclusion

Basic (structured) prompts are more effective than naïve prompts in generating high-quality, accurate, and detailed responses. While naïve prompts may work for simple queries, structured prompts are essential for complex and academic tasks.

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8. Recommendations for Writing Better Prompts

- Be **specific and clear**
- Provide **context**
- Define **output format**
- Mention **length or constraints**
- Include **examples if needed**